

Branching Logic

Project: Transparent Book Recommender

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Decompose a Goal

Without opening the old lesson, explain one key idea from Decompose a Goal. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Trace the tested path for Branching and boundary tests. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Branching and boundary tests. Proposed explanation: “Every condition in the chain always runs”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Build a new sorter independently

Use an integer tiles count. Label a negative value Invalid, zero Empty, 1 through 6 Small and values above 6 Large. Write an ordered if elif else script and predict -1, 0, 1, 6 and 7. Include one print after the chain.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

Classify lengths with if pages<80 then Short; else if pages<=160 then Medium; else Long. Trace 79,80,160,161. What happens if the <=160 branch is placed first?

6. Solve the given case

If pages<100 choose Short; otherwise if pages<=200 choose Medium; otherwise Long.

Classify 99,100,200,201 pages.

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

If pages are at most 80, choose short; otherwise choose long. Classify books with 79, 80 and 81 pages and explain the boundary.

My independent answer and reason

Write or draw your reasoning and evidence.

Week 3 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Branch when a reader has a hard constraint such as language, format, or maximum length.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

Classify lengths with if pages<80 then Short; else if pages<=160 then Medium; else Long. Trace 79,80,160,161. What happens if the <=160 branch is placed first?

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Explain the chain

Describe the purpose of if, elif and else in the worked duration sorter. Explain why print(label) is outside the indented branches.

My reasoning and evidence

13. W2 Trace the tested path

For -1, 12 and 13, record checked conditions, skipped conditions and the final label. Run each as a fresh script after predicting.

My reasoning and evidence

14. W3 Repair the overwritten result

Trace the two-independent-if example for -1. Identify which assignment overwrites the earlier label and make the smallest repair.

My reasoning and evidence

15. W4 Build a new sorter independently

Use an integer tiles count. Label a negative value Invalid, zero Empty, 1 through 6 Small and values above 6 Large. Write an ordered if elif else script and predict -1, 0, 1, 6 and 7. Include one print after the chain.

My reasoning and evidence

16. W5 Choose revealing regression tests

A repair changes the small-count condition from < 6 to ≤ 6 . Name the case that exposes the original error and two different kinds of earlier cases to rerun.

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: Branching Logic

A fictional reading-time labeler uses one Python if/elif/else chain.

Whole minutes below 0: Invalid; 0: Empty; 1–10: Short; above 10: Long.

Order checks as `minutes < 0`, `minutes == 0`, `minutes <= 10`, otherwise. Print the assigned label once.

AI ACADEMY / AI-2/3

Branching Logic

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Negative check

-1 takes Invalid and skips later branches.



2. Zero check

0 takes Empty, not Short.



3. Inclusive short range

1 and 10 take Short.



4. Remaining inputs

11 takes Long; only the selected branch assigns the label.

CHANGE ONE THING

Move `minutes <= 10` to the first branch.

Explain the changed result using the matching lesson.

Give expected labels for -1, 0, 10, 11 and identify the equality boundary for Short.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: Validate recommender requests before ranking items, retaining explicit labels for missing or invalid cases.